Fully Contextual Network for Hyperspectral Scene Parsing

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***Abstract*— In this article, we propose fully contextual networks (FullyContNets) for hyperspectral scene parsing. Different from the previous approaches that leveraging the local information, the proposed methods can effectively capture the more generic nonlocal contexts. To this end, we first propose the scale attention module (SAM) that can adaptively aggregate the multiple features through obtaining the interfeature dependencies of multiscale with self-attention mechanism, where the weights are determined by measuring the similarity between features. What is more, two fully contextual modules (FCMs) called pyramid fully contextual module (Pyramid-FCM) and atrous spatial pyramid fully contextual module (ASP-FCM) are separately developed to obtain the contextual information that simultaneously lying across positions, channels, and features when combining the intrafeature information aggregation algorithms with SAM on the foundation of existing multiscale modules, such as pyramid pooling (PP) in PSPNet and atrous spatial pyramid pooling (ASPP) in DeeplabV3. We design four schemes for FCMs to obtain more effective contexts. The corresponding FullyContNet-Pyramid and FullyContNet-ASP are separately constructed based on the Pyramid-FCM or ASP-FCM. There are extensive quantitative and qualitative experiments are conducted, depicting the capability of SAM and FCMs and demonstrating the competitiveness of proposed networks on four public hyperspectral scenes when comparing with the current state-of-the-art approaches.**

***Index Terms*— Adaptively, fully contextual network (FullyContNet), hyperspectral scene parsing, nonlocal, scale attention module (SAM).**

I. Introduction

With the development of earth observation technology, the hyperspectral image (HSI), which usually possesses hundreds of continuous and narrow bands and contains abundant spectral and spatial information, is now available and plays a key role in many applications, including agriculture [1], environmental management [2], and anomaly detection [3]. Among these fields, the hyperspectral scene parsing, which is also known as the HSI classification, has been conducted as a fundamental task that aims to assign a certain category label to the pixels of the whole image and is receiving considerable interest over the past years. Note that the HSI classification and hyperspectral scene parsing are two different implementations of the same task; the former focus on classification pixel by pixel, thus usually using the classification model, while the latter is more inclined to the segmentation of the whole image. Thus, in this article, we adopt the concept of "hyperspectral scene parsing" rather than "HSI classification" since the proposed methods, which will be introduced later, are image-level interpreting models that can simultaneously predict the categories of all pixels. However, this topic is still challenging since the different factors, such as scales, shapes, and locations of the various objects, interfere with the judgment of the pixel categories.

The early methods, such as k-nearest neighbors [4], multinomial logistic regression [5], random forest [6], and support vector machine (SVM) [7], are only based on spectral information since they classify the HSI only with the spectral features, which are produced by preprocessing the HSI through the principal component analysis [8], the independent component analysis [9], or the linear discriminant analysis [10]. However, the accuracies of these methods are limited due to lacking surrounding information. To address this problem and further improve the classification, spatial information is incorporated into consideration. For example, the extended morphological profiles [11], the gray-level cooccurrence matrix [12], the superpixel segmentation [13], and multiple kernel learning [14], [15] are involved to generate discriminative spectral–spatial features for classification, where the SVM [16] is the most commonly used classifier because of its stable performance in processing high-dimensional data. Thus, the SVM is served as the baseline method in the later performance comparison experiments. Besides these feature engineering approaches that improving the features before learning a classifier, there are some other solutions to tackle this issue. Concretely, they refine the preliminary classification results by means of the Markov random field [17] or edge-preserving filtering [18]. In addition, some other methods based on sparse representation [19], active learning [20], and wavelet transform [21] have also been exploited.

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However, although the above methods perform well, they usually generate the handcrafted features that are extracted by the designed specific strategy. Therefore, their performance strongly depends on the experience of the designer. In addition, these features are not robust in the complicated environments [22] since they are regarded as the shallow features, which cannot characterize the essential attributes of the objects because the corresponding methods belong to the shallow-layer model [23]. To obtain the more generic features, the data-driven deep learning technology is widely applied into HSI classification, where the fully connected network is the earliest employed [24], [25]. However, these methods restrict the size of input data that all multidimensional data need to be flattened to 1-D; thus, the spatial structure is inevitably lost. To alleviate this issue, with the help of convolutional neural networks (CNNs), which can simultaneously implement the local perception and maintain the spatial structure, many methods are introduced into HSI classification [26]–[29]. Furthermore, the recurrent neural network (RNN) and its variants, including long short-term memory (LSTM) and gated recurrent units, are also used to extract the spectral–spatial contextual information [30]–[33].

Many spectral–spatial deep learning methods still follow the patch-level classification pipeline: generating the patches that are the neighbor regions within a certain range around the selected samples and feeding these patches into the network so as to generate a category label. However, it means that unnecessary computations are brought because adjacent patches usually have overlapping area, causing redundant training. What is more, the limited size of input patches not only restricts the receptive field of the network but hinders the modeling of long-range dependence. In fact, except for the abundant spectral channels, the format of input data and ground truth in HSI are very similar with the natural image, indicating that the parsing of HSI can follow the similar approaches in the natural image, where the fully convolutional networks (FCNs) have been extensively employed. Up until now, the dual-path FCN [34] and encoder–decoder architecture [35] have been separately applied into the parsing of HSI. However, these networks cannot fully exploit the contexts since the receptive fields are increased still only by stacking convolutional layers and the nonlocal contexts cannot be captured because atrous convolution and ordinary convolution all belong to a local operator. Then, the spatial nonlocal self-attention modules are utilized by some researchers to extract contextual information in a larger range [36]–[38]. Despite this, the full contexts have not been explored, and it is still has room for improvement because the nonlocal modules that they use can only capture the contexts of different pixels within a certain feature, while the contextual information across multiple features, especially the multiscale features, which are effective for hyperspectral scene parsing [39]–[41], has not been obtained.

In fact, the dependence among different scales clearly exists: the large objects are more easily captured with the broad view features, while the small objects or boundaries are tended to be maintained by narrow range information, suggesting that different attentions should be drawn not only on various area of the feature at a specific scale but also on different scales of a certain region. Our motivation is to obtain the interfeature dependencies of multiple scales and bridge the gap of themselves. Therefore, in this article, based on the self-attention mechanism, we propose a scale attention module (SAM) that can adaptively capture the dependencies between the feature in a certain scale and other scale features through measuring the correlation in mapped latent feature space to depict the different scale importance in a specific region. What is more, if the interfeature contexts capturing module SAM is combined with the intrafeature position aggregation modules (PAM), the richer contexts that simultaneously across scales and spaces can be obtained. Fig. 1 shows the difference of the context capturing extent whether the interfeature dependencies exist. It should note that the SA is space-saving because it does not require multiple inputs. The complexity of SA is $O(N_s \times N_s)$, where N_s is the number of scales, which is usually a relatively small value compared with feature size or channel number.

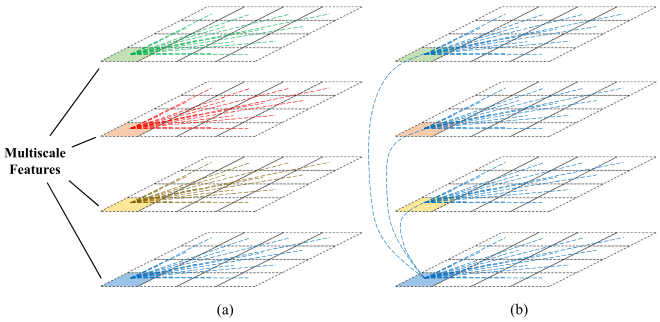

Fig. 1. Illustration of the context-dependence capturing extent for the blue pixel. (a) Only intrafeature context-aware. (b) With the interfeature context-aware. The perception scope of the pixel is represented by a dotted line with the same color. It can be seen that the context capturing the extent of the blue pixel expands to other features after building the interfeature dependence.

Besides the scale and spatial position, to achieve fully contextual capturing, the channel also needs to be considered. As the FreeNet [35] shown, the importance of different channel is essential to be readjusted, while the effect of corresponding channel attention module (CAM) has been presented in DANet [36]. In this article, PAM, CAM, and the proposed SAM that separately capture the contexts in various aspects are integrated together to fully perceive the contextual information. Concretely, considering the PAM and CAM perform well on high-level semantics, meaning that the generated multiscale features, which is prepared for SAM, should also belong to the high-level features. Thus, two widely used multiscale modules "pyramid pooling" (PP) in PSPNet [42] and "atrous spatial pyramid pooling" (ASPP) in DeeplabV3 [43] are directly applied on top features and reasonably combined with the above mentioned three context capturing modules, producing two extended fully contextual modules (FCMs): "pyramid fully contextual module" (Pyramid-FCM) and "atrous spatial pyramid fully contextual module" (ASP-FCM). Furthermore, based on these modules, the corresponding networks, FullyContNet-Pyramid and FullyContNet-ASP, are separately developed for hyperspectral scene parsing, and these networks all can learn in an end-to-end manner. The name "FullyContNet" is the abbreviation of "fully contextual network"

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so as to distinguish with "FCN" for convenience. The main contributions of this article can be summarized as follows.

1) We propose an SAM that can effectively bridge the interfeature gaps and adaptively capture the dependencies lying across multiscale features.
2) We develop the FCMs that can perceive the contexts simultaneously across positions, channels, and features based on the proposed SAM using the multiscale features generated by PP and ASPP. In addition, we design different schemes of FCMs to achieve better accuracies.
3) We construct the fully contextual networks named FullyContNet for hyperspectral scene parsing taking advantage of the FCMs. Extensive quantitative and qualitative experiments demonstrate that the proposed methods perform to be competitive on four hyperspectral scene parsing benchmarks compared with other state-of-the-art methods.

The remainder of this article is organized as follows. Section II gives a brief introduction of related works. Section III describes the proposed FCMs and networks. Experiments and related comprehensive analysis are presented in Section IV. Finally, Section V concludes this article.

II. Related Work

A. Natural Scene Parsing

Since the FCNs are used by FCN for dense prediction tasks, such as scene parsing or semantic segmentation into the natural scene, many FCN-based methods keep making great progress. UNet [44], SegNet [45], DeconvNet [46], RefineNet [47], and discriminative feature network (DFNet) [48] adopt encoder–decoder architectures to carefully recover the details in the upsampling procedure. To perceive the global information, global convolutional network (GCN) [49] uses the larger kernel, and BiSeNet [50] adopts the global pooling, while an effective encoding layer is introduced in EncNet [51] on top of the network. There are also some networks extracting the features from the perspective of categories [52], [53]. In the aspect of multiscale feature generating, multiple atrous convolution filters are employed in DeeplabV3 [43] to construct the ASPP module so as to regularly capture the objects in different ranges of the target pixel. DenseASPP [54] further increases the receptive field with the dense connection, while PSPNet parallel meshes the feature map to different size using the PP module. The Deeplab-field of vision (FOV)-Attention [55] and Hierarchical-Multiscale-Attention [56] obtain the multiscale features through different size inputs with multiple backbone networks and then combine these features with the attention model. We must argue that our SAM is totally different from them since they combine the multiscale features through weighted addition by using the explicitly generated attention map, while we adopt a self-attention mechanism to capture the dependence of interfeatures for multiscale contexts aggregation, where the attention map is produced implicitly.

B. Hyperspectral Scene Parsing

As far as we know, there are only a few studies studying the hyperspectral scene parsing. The first one involved in this field is spectral–spatial fully convolutional network (SSFCN) [34], which is exactly a double-branch network to separately extract spectral and spatial features, where the atrous convolutions are used to expand the vision field. Then, the FreeNet [35], an encoder–decoder architecture network, is developed, where the squeeze-and-excitation block [57] is employed to improve the classification. Following feature pyramid network (FPN) [58], FreeNet adopts the lateral connections in the upsampling procedure to gradually restore the image resolution. With the breakthrough progress of the self-attention mechanism (introduced next) in the computer vision field, to achieve the nonlocal contexts capturing on HSI, ENL-FCN [38] parallel uses two recurrent criss-cross attention (RCCA) modules, which is proposed in CCNet [37], to perceive the contextual information in a wider scope. Different from the abovementioned works, regarding the scale contexts lying in multiple features, we propose a nonlocal module named SAM that aims to perceive the interfeatures dependence and further present a deep network called FullyContNet that can capture the more effective fully contextual information, which simultaneously across positions, channels, and features, so as to assist the hyperspectral scene parsing.

C. Self-Attention Mechanism

The self-attention mechanism is first proposed for machine translation in [59] and then applied in the computer vision field through the nonlocal module [60], which overcomes the drawback of the previous local perception since it can capture the long-range dependence and provide a broader view for any position by computing the correlation between spatial points in the mapped latent semantic space. DANet [36] and OCNet [61] use the nonlocal module to capture the spatial context, while PSANet [62] aggregates the information with the attention maps that adaptively generated by two branches. In order to reduce the computational complexity of the nonlocal module, CCNet [37] recurrently leverages the pixels in a criss-cross region that is in the same row or column as the target pixel. Different from the above methods that conducted within features, our SAM can perceive the relationship between multiscale features, while the FCM can further simultaneously capture the nonlocal dependencies across regions, channels, and features.

The unified attention model includes three parts: key, query, and value [63]. Assume that there are T groups keys, queries, and values $\langle k_i, q_i, v_i \rangle$, $i = 1, 2, \ldots, T$. First, evaluating the similarity between the ith query q_i and each key, in turn, and obtaining the energy score

$$e_{ij} = f(q_i, k_j). \tag{1}$$

Before weighting the corresponding values, the energy score need to be normalized using the softmax function

$$\hat{e}_{ij} = \text{softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{j=1}^{T} \exp(e_{ij})}. \tag{2}$$

The corresponding ith result r_i is obtained by weighted adding the values

$$r_i = \sum_{j=1}^{T} \hat{e}_{ij} \cdot v_j. \tag{3}$$

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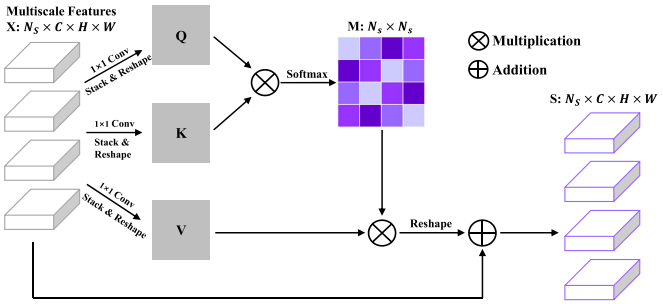

Fig. 2. Details of SAM.

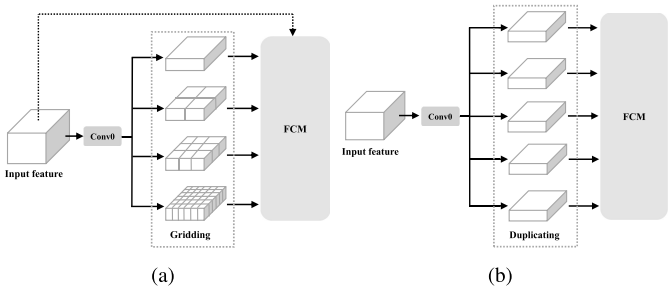

Fig. 3. Proposed FCMs. (a) Pyramid-FCM. (b) ASP-FCM. The dotted line is residual skip connection of the input feature in Pyramid-FCM.

The whole procedure can be formulated as

$$r = \text{Attention}(K, Q, V) = \text{softmax}(QK^T)V \tag{4}$$

where K, Q, and V are the packed matrices of keys, queries, and values.

The attention model becomes a self-attention mechanism if the key, query, and value are the same variable, which is the critical technology of the proposed methods.

III. Proposed Methods

In this section, we first introduce the proposed SAM. Then, we give the details of the Pyramid-FCM and ASP-FCM. The architecture of FullyContNet will be presented in the end.

A. Scale Attention Module

Previous context aggregation modules, such as PAM and CAM, always operate within the feature, while the proposed SAM can capture the contextual information across the multiscale features. Given a group of feature maps in the same size $\mathbf{X} = \{X_i \in \mathbb{R}^{C\times H\times W}\}_{i=1}^{N_s}$, where H, W, and C are the height, width, and channel number of these feature maps, and N_s is the scale number. Fig. 2 shows the details of SAM. Concretely, the SAM maps the input multiscale features $\mathbf{X}$ to a new group $\mathbf{S} = \{S_i \in \mathbb{R}^{C\times H\times W}\}_{i=1}^{N_s}$, which collects the interfeature contextual information of $\mathbf{X}$

$$\mathbf{X} \xrightarrow{\text{SAM}} \mathbf{S}. \tag{5}$$

Similar to the self-attention mechanism, in SAM, the input multiscale features $\mathbf{X}$ are first transformed to keys and queries, which are symbolized as $\mathbf{K}$ and $\mathbf{Q}$ through a 1×1 convolution layer followed by stacking and reshaping, respectively. Each feature map X_i in $\mathbf{X}$ should pass the 1×1 convolution and then stack with other transformed features. It should be noted that the stacking is to concatenate the feature maps at the scale dimension. The obtained $\mathbf{Q}$ and $\mathbf{K}$ are in size of $\mathbb{R}^{N_s\times N}$, where $N = C' \times H \times W$, and C' is the channel number after 1×1 convolution. Then, we evaluate the similarity between different element pairs in $\mathbf{Q}$ and $\mathbf{K}$ as the formula (1). In fact, the $f(\cdot)$ in (1) is implemented as the inner product. Thus, after matrix multiplication $\mathbf{QK}^T$ and softmax function, we obtain the SA map $\mathbf{M} \in \mathbb{R}^{N_s\times N_s}$

$$\mathbf{M}_{ij} = \text{softmax}(\mathbf{Q}_i\mathbf{K}_j) = \frac{\mathbf{Q}_i\mathbf{K}_j}{\sum_{j=1}^{N_s}\mathbf{Q}_i\mathbf{K}_j}. \tag{6}$$

At the same time, the corresponding values $\mathbf{V} \in \mathbb{R}^{N_s\times CHW}$ of $\mathbf{X}$ are also generated in the same way as obtaining $\mathbf{K}, \mathbf{Q}$ except that there are different channel number of 1×1 convolution layer in which is the same as $\mathbf{X}$. Then, the same as (3), we regroup the values according to the SA map $\mathbf{S}$. In addition, following [36], we multiply a scale parameter λ to the reshaped aggregation result and implement an elementwise addition with the initial features $\mathbf{X}$; this procedure is

$$\mathbf{S}_i = \lambda \cdot \text{reshape}(\mathbf{M}_i\mathbf{V}) + \mathbf{X}_i \tag{7}$$

where $\mathbf{S} \in \mathbb{R}^{N_s\times C\times H\times W}$, and λ can be learned adaptively through end-to-end training. From the above procedure, it can be seen that each feature S_i in $\mathbf{S}$ reasonably captures the contexts of other scale features. Hence, the SAM can selectively perceive and aggregate contextual information across features. What is more, it constructs the connection across features and bridges the semantic gap between different scales, laying the foundation for more comprehensive context-awareness in the next FCMs.

B. Fully Contextual Module

The SAM only captures the context dependencies across multiscale features. A more general idea is to perceive the contextual information that simultaneously lying across the region, channel, and feature. Therefore, if the proposed SAM is combined with the intrafeature information aggregation algorithms, such as PAM and CAM, we can obtain more full contexts of multiple scales, spatial positions, and bands.

To this end, it is essential to generate multiscale features. Thus, we directly utilize two universal and commonly used modules: PP and ASPP, where the multiple scales are separately determined by different settings in the parallel paths, for example, the grid pooling sizes in PP and dilation rates of the atrous convolution in ASPP. The proposed Pyramid-FCM and ASP-FCM are shown in Fig. 3. A residual skip connection that the input feature prepared for combining with the output of the subsequent FCM to maintain the original information is possessed in the Pyramid-FCM, while, in ASP-FCM, before feeding into the FCM, the input feature is dimensionally reduced with a 1×1 convolutional layer Conv0 to accelerate the computation. In Pyramid-FCM, the features are separately aggregated in each grid using PAM in the subsequent FCM part. Thus, before feeding into FCM, we first generate the grids, which are in fact the virtual boundaries that restricting

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the scope of PAM, according to the different setting of gridding pooling size in each path. While, in ASP-FCM, the atrous convolutions and PAM are all implemented in FCM. Thus, before FCM, the feature after Conv0 is merely duplicated five times.

It is natural to consider that contextual information that simultaneously lying across features and regions could be perceived if the contexts of interfeature are captured after obtaining the intrafeature information. Assume that there is a group of multiscale features $\mathbf{X} = \{X_i \in \mathbb{R}^{C\times H\times W}\}_{i=1}^{N_s}$. First, we directly utilize the PAM to capture the spatial dependence within features and obtain the intermediate output $\mathbf{P} = \{P_i \in \mathbb{R}^{C\times H\times W}\}_{i=1}^{N_s}$. Concretely, the algorithm is parallel applied on each scale feature X_i

$$X_i \xrightarrow{\text{PAM}} P_i. \tag{8}$$

In P_i, assume that the jth pixel $p_{i,j}$ obtains the information of all the pixels in X_i; thus

$$p_{i,j} = w_{i,:}^{p_j} \bullet x_{i,:} \tag{9}$$

where $w_{i,:}^{p_j} \in \mathbb{R}^{1\times HW}$ is the position attention vector of the jth pixel in X_i, and $x_{i,:} \in \mathbb{R}^{HW\times C}$ is all the pixels of X_i, while "$\bullet$" is the matrix multiplication. Then, we adopt SAM to capture the contexts in features $\mathbf{P}$ and obtain $\mathbf{S} = \{S_i \in \mathbb{R}^{C\times H\times W}\}_{i=1}^{N_s}$; this procedure can be formulated as

$$\mathbf{P} \xrightarrow{\text{SAM}} \mathbf{S}. \tag{10}$$

For $p_{i,j}$, SAM can obtain the contextual information in other features at the same position, so

$$s_{i,j} = w_{:,j}^{s_i} \bullet p_{:,j} \tag{11}$$

where $w_{:,j}^{s_i} \in \mathbb{R}^{1\times N_s}$ is the SA vector of P_i in position j, and $p_{:,j} \in \mathbb{R}^{N_s\times C}$ is the jth pixel in each P_i of $\mathbf{P}$.

Combining the formula (9) and (11), $s_{i,j}$ can be expanded as

$$\begin{aligned} s_{i,j} &= w_{:,j}^{s_i} \bullet p_{:,j} \\ &= \sum_{m=1}^{N_s} w_{m,j}^{s_i} \cdot p_{m,j} \\ &= \sum_{m=1}^{N_s} w_{m,j}^{s_i} \sum_{n=1}^{HW} \left(w_{m,n}^{p_j} \cdot x_{m,n}\right) \\ &= \sum_{m=1}^{N_s} \sum_{n=1}^{HW} \left(w_{m,j}^{s_i} \cdot w_{m,n}^{p_j}\right) \cdot x_{m,n}. \end{aligned} \tag{12}$$

In fact, in SAM, $w_{m,j}^{s_i} = w_{m,k}^{s_i} = w_m^{s_i}, k = 1, \ldots, HW$; thus, the above formula is finally reduced to

$$s_{i,j} = \sum_{m=1}^{N_s} \sum_{n=1}^{HW} \left(w_m^{s_i} \cdot w_{m,n}^{p_j}\right) \cdot x_{m,n} \tag{13}$$

where $w_m^{s_i}$ and $w_{m,n}^{p_j}$ can be adaptively learned by the network in training procedure, indicating that any pixel in an arbitrary feature can aware the contextual information of all the other pixels regardless of whether they are on the same feature after contexts capturing using PAM and SAM so as to construct the more generic and broader dependencies that are simultaneously lying across positions and features.

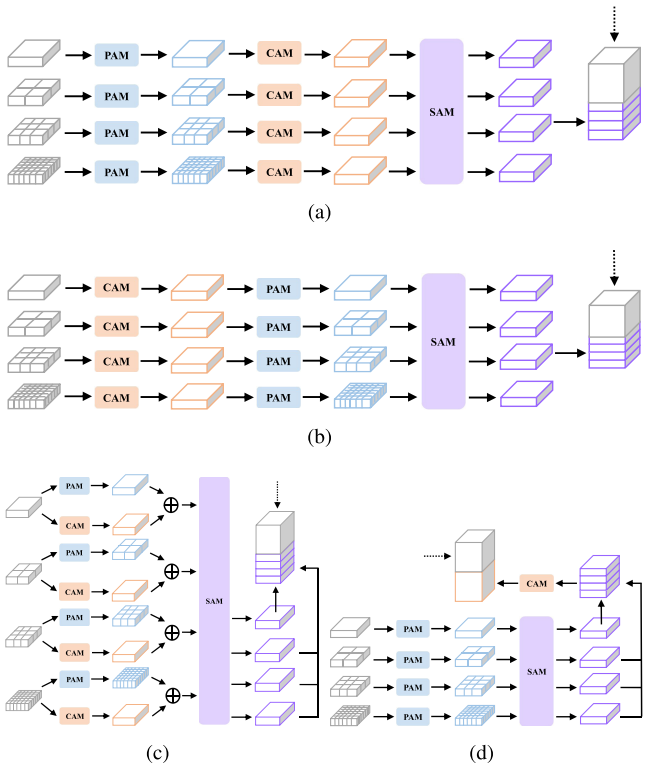

Fig. 4. Four schemes of Pyramid-FCM. (a) P-C-S. (b) C-P-S. (c) P + C-S. (d) P-S-C. The dotted line is residual skip connection of the input feature in Pyramid-FCM.

However, the sequential of PAM and SAM cannot be exchanged. If we implement SAM before PAM, now

$$\begin{aligned} s_{i,j} &= w_{:,j}^{s_i} \bullet x_{:,j} \\ p_{i,j} &= w_{i,:}^{p_j} \bullet s_{i,:}. \end{aligned} \tag{14}$$

Expanding $p_{i,j}$ can get

$$\begin{aligned} p_{i,j} &= w_{i,:}^{p_j} \bullet s_{i,:} \\ &= \sum_{n=1}^{HW} w_{i,n}^{p_j} \cdot s_{i,n} \\ &= \sum_{n=1}^{HW} w_{i,n}^{p_j} \sum_{m=1}^{N_s} \left(w_{m,n}^{s_i} \cdot x_{m,n}\right). \end{aligned} \tag{15}$$

$w_{k,n}^{p_j}, k \neq i$ cannot be perceived by $p_{i,j}$ because the PAMs are parallel conducted on each feature. The comparison of formula (12) and (15) demonstrates the effect of SAM that serving as a bridge to associate different features.

Therefore, if you want to obtain the position and scale contextual information at the same time, the SAM must conduct after the PAM, while the CAM can be arranged in any sequence because it mainly used to improve the feature representations [36]. According to this principle, we design four schemes for the proposed FCM to obtain the context dependencies as comprehensive as possible. Figs. 4 and 5 separately depict these schemes of Pyramid-FCM and ASP-FCM, where "P" represents the position contexts obtaining using PAM. In this article, considering the high computing

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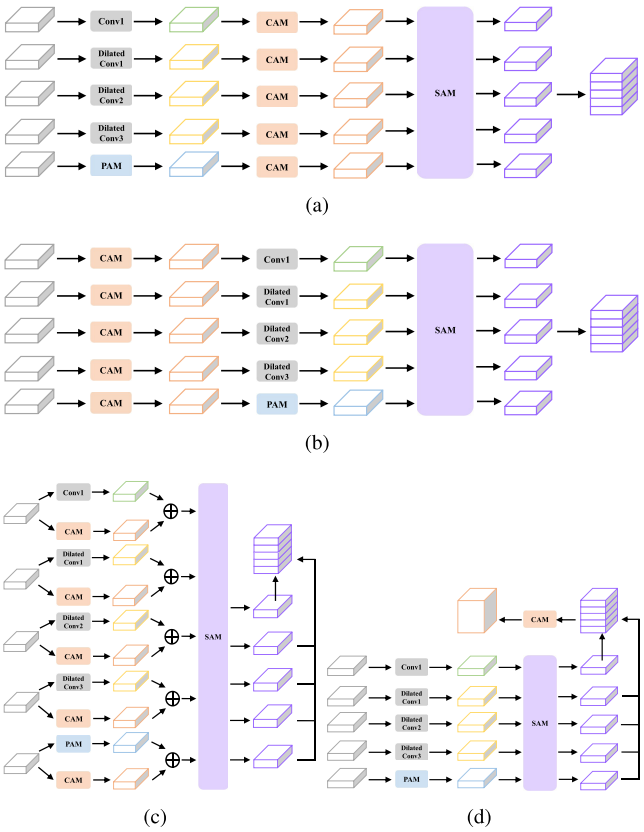

Fig. 5. Four schemes of ASP-FCM. (a) P-C-S. (b) C-P-S. (c) P + C-S. (d) P-S-C.

complexity of nonlocal module, we choose the RCCA proposed in CCNet [37] as the PAM, and the recurrent number is set to 2 for fully aggregating the spatial information. The contexts across channels and scales can be captured with CAM and SAM that are marked as "C" and "S," respectively. "-" is the meaning of sequential, while "+" is the addition of two parallel branches. The details of these schemes are introduced next.

1) *P-C-S:* First, obtaining position contexts within the feature using PAM. Then, the CAM is used to improve the single feature through readjusting channels. Finally, the proposed SAM is applied to capture the interfeature context dependence for fully contextual information perception. In Pyramid-FCM, the PAM and CAM are separately employed in each path to obtain the multiscale features. Noted the input feature of Pyramid-FCM is split with grids. The multiple scale position contexts are generated when the context capturing of any pixels in each path is limited to the grids because these paths have different grid sizes. The schematic of the PAM using in a single path has been shown in Fig. 6. The setting of grid division follows the original PP module in PSPNet [42]. In ASP-FCM, the original global average pooling in ASPP [43] is replaced with a PAM to more flexibly capture the global context. The convolutional layers (Conv1 and Dilated Conv1-3, shown in green or yellow) are kept since they have different dilation rates and have the ability to perceive the contexts of various ranges to produce the multiscale features, playing a similar role as PAM (same below).
2) *C-P-S:* The CAM is first used to capture the interdependencies among different bands. Then, the contexts of spatial and multiple scales are extracted to obtain fully contextual information. Compared with the "P-C-S" scheme, as shown in Figs. 4(b) and 5(b), this scheme only exchanges the order of position and channel contexts acquiring.
3) *P + C-S:* Similar with DANet [36], different with the sequential connection of PAM and CAM, in each path of Pyramid-FCM and ASP-FCM, this scheme adopts two branches to parallel extract position and channel contexts and adds the results together. The SAM is used to obtain the interfeature dependence in the end.
4) *P-S-C:* Different from the above three schemes, this scheme treats the feature after position and scale contexts capturing as a whole and places the CAM in the end. Therefore, before the SAM, only the position information of the feature in each path needs to be aggregated.

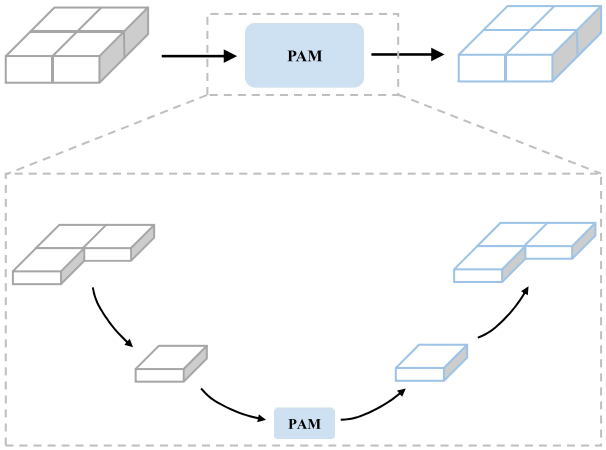

Fig. 6. PAM using in each path of the Pyramid-FCM.

After SAM, the multiscale features, which have abundant contexts of various positions and channels and even obtain the dependence of other features, are concatenated in channel dimension. Except that the "P-S-C" scheme needs to process the fused feature using CAM, other schemes have so far finished the extraction of the full contexts. Moreover, in Pyramid-FCM, the obtained contexts are further combined with the original input that provides the prior information [42]. Finally, the information of concatenated feature is integrated by reducing the dimension with a 1×1 convolutional layer named Conv-DR1 followed by a group normalization (GN) [64] layer and the rectified linear unit (ReLU) activation function as the batch size always equals to 1.

C. Fully Contextual Network

Fig. 7 shows the architecture of the proposed FullyContNet. Different from the previous patch-based methods, that is, classification with the local information, which is obtained in a fixed size around the target pixel, our method directly

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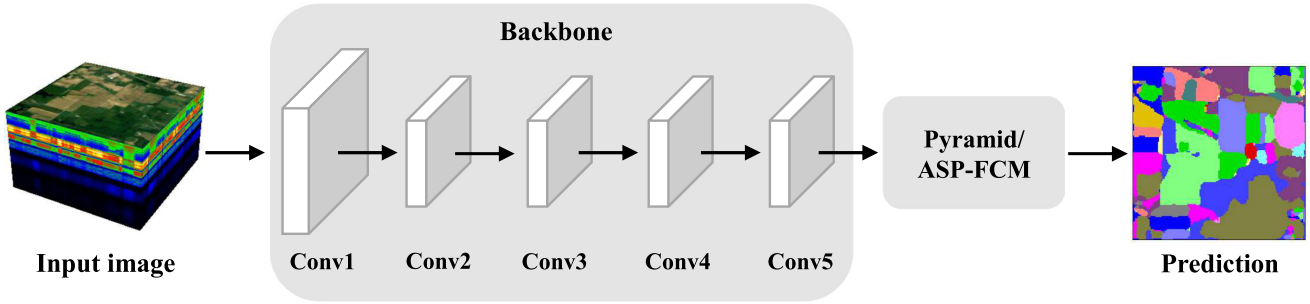


Fig. 7. Architecture of the proposed FullyContNet.

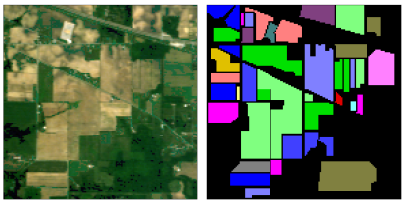
feeds the whole image into the trained network and produces the scene parsing result. The FullyContNet includes two parts: backbone and FCM; the former is an FCN that consists of five 3×3 convolutional layers, while the latter is Pyramid-FCM or ASP-FCM. It should be noticed that, in this article, the convolutional layers are not necessarily a single convolution since they may be configured as a composite layer that contains multiple operations, whose structure and parameter details will be shown later. First, a convolutional layer Conv1 is used to preliminarily encode the input image. Before feeding into the next convolutional layer, the output of Conv1 is downsampled with average pooling to save computing resources. The current resolution is maintained until through the FCM part. Finally, before classification, the output of Pyramid-FCM or ASP-FCM has further conducted dimension reduction with a 1×1 convolutional layer called Conv-DR2, and the resolution is recovered to the initial size using bilinear interpolation. It should note that we revise the dilation rates of the proposed ASP-FCM compared with the original ASPP module according to the setting of "output stride = 8," which is a common configuration of a series of scene parsing networks for natural image based on dilated FCN, because the input HSI are in various sizes. Specifically, the dilation rates can be adaptively determined by the size of HSI since we find that the maximum theoretical receptive field length of the dilated convolution layers (Dilated Conv3) in the original ASPP module is about 75% of the input feature size. Therefore, in our method, the maximum dilation rate d_3 of Dilated Conv3 is

$$d_3 = \lfloor [\min(h, w)/2] * 0.75/2 \rfloor \tag{16}$$

where h and w are the height and width of input HSI, and "$\lfloor \rfloor$" is the rounding down operation. Similar to the setting of original ASPP module, the dilation rates of other dilated convolution (Dilated Conv1-2) are then can be computed as $d_1 = \lfloor (d_3/4) \rfloor, d_2 = \lfloor (d_3/2) \rfloor$. All 3×3 convolutional operations are followed by a GN layer and the ReLU function.

IV. Experiments

In this section, we first introduce the implementation details and data sets, and then, we conduct a series of qualitative and quantitative comprehensive analyses. In order to more convincingly evaluate the effectiveness of the proposed methods, we choose four hyperspectral data sets, where two rural scenes, called Indian Pines and Salinas Valley, and two urban scenes, named Pavia University and the University of Houston, are included.



(a) (b)

ID	Color	Category	Trn.	Val.	Tes.	Tot.
1		Alfalfa	26	7	13	46
2		Corn-notill	80	20	1328	1428
3		Corn-mintill	80	20	730	830
4		Corn	80	20	137	237
5		Grass-pasture	80	20	383	483
6		Grass-trees	80	20	630	730
7		Grass-pasture-mowed	16	4	8	28
8		Hay-windrowed	80	20	378	478
9		Oats	11	3	6	20
10		Soybean-notill	80	20	872	972
11		Soybean-mintill	80	20	2355	2455
12		Soybean-clean	80	20	493	593
13		Wheat	80	20	105	205
14		Woods	80	20	1165	1265
15		Buildings-Grass-Trees-Drives	80	20	286	386
16		Stone-Steel-Towers	60	15	18	93
Tot.			1073	269	8907	10249

(c)

Fig. 8. Indian Pines data set. (a) False-color image. (b) Ground truth. (c) Category and sample settings (the number of training, validation, testing, and total samples are separately abbreviated as "Trn.,""Val.," "Tes.," and "Tot.").

A. Data Set

1) Rural Scene: Indian Pines: This scene was gathered at Northwestern Indiana by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992, consisting of 200 bands with a size of 145×145 pixels that, in 20 m, spatial resolutions after water absorption bands were removed and in the wavelength range of 0.4–2.5 μm; 16 vegetation classes are involved in this scene, and related information such as original image visualization, ground truth, and configuration of training, validation, and testing samples has been presented in Fig. 8.

2) Rural Scene: Salinas Valley: This scene was collected by the AVIRIS sensor over Salinas Valley, California, consisting of 204 bands with a size of 217×512 samples in a 3.7-m spatial resolution. As with the Indian Pines scene, 20 water absorption bands were discarded. It contains 16 classes, including vegetables, bare soils, and vineyard fields. The related information is shown in Fig. 9.

3) Urban Scene: Pavia University: This scene was obtained over Pavia University at Northern Italy by the Reflective Optics System Imaging Spectrometer (ROSIS) in 2001, consisting of 103 bands with a size of 610×340 pixels in 1.3-m spatial resolution and in the wavelength range of 0.43–0.86 μm; nine categories are included in this data, which is shown in Fig. 10.

4) Urban Scene: University of Houston: This scene was acquired over the University of Houston campus and its

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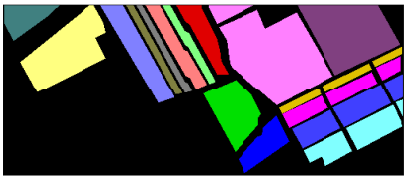
(a)

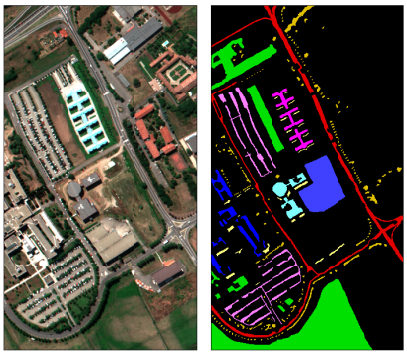


(b)

ID	Color	Category	Trn.	Val.	Tes.	Tot.
1		Brocoli green weeds 1	80	20	1909	2009
2		Brocoli green weeds 2	80	20	3626	3726
3		Fallow	80	20	1876	1976
4		Fallow rough plow	80	20	1294	1394
5		Fallow smooth	80	20	2578	2678
6		Stubble	80	20	3859	3959
7		Celery	80	20	3479	3579
8		Grapes untrained	80	20	11171	11271
9		Soil vinyard develop	80	20	6103	6203
10		Corn senesced green weeds	80	20	3178	3278
11		Lettuce romaine 4wk	80	20	968	1068
12		Lettuce romaine 5wk	80	20	1827	1927
13		Lettuce romaine 6wk	80	20	816	916
14		Lettuce romaine 7wk	80	20	970	1070
15		Vinyard untrained	80	20	7168	7268
16		Vinyard vertical trellis	80	20	1707	1807
Tot.			1280	320	52529	54129

(c)

Fig. 9. Salinas Valley data set. (a) False-color image. (b) Ground truth. (c) Category and sample settings.



(a) (b)

ID	Color	Category	Trn.	Val.	Tes.	Tot.
1		Asphalt	80	20	6531	6631
2		Meadows	80	20	18549	18649
3		Gravel	80	20	1999	2099
4		Trees	80	20	2964	3064
5		Metal sheets	80	20	1245	1345
6		Bare soil	80	20	4929	5029
7		Bitumen	80	20	1230	1330
8		Bricks	80	20	3582	3682
9		Shadows	80	20	847	947
Tot.			720	180	41876	42776

(c)

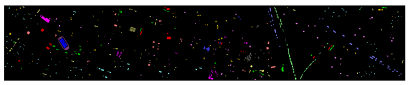
Fig. 10. Pavia University data set. (a) False-color image. (b) Ground truth. (c) Category and sample settings.

neighbor regions by ITRES-CASI 1500 sensor in 2012, containing 144 bands with a size of 349 × 1905 pixels in a 2.5-m spatial resolution and in the wavelength range of 0.4–1.0 μm; 15 categories are included in this data, which is depicted in Fig. 11. This scene was also used in the 2013 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest.

B. Implementation Details and Experimental Settings

The methods are implemented with Pytorch. The base learning rate is set to 0.01, which is adjusted with the poly scheduling policy and multiplied by $(1 - (\text{iter}/\text{max_iter}))^{0.9}$.

(a)



(b)

ID	Color	Category	Trn.	Val.	Tes.	Tot.
1		Grass healthy	80	20	1151	1251
2		Grass stressed	80	20	1154	1254
3		Grass synthetic	80	20	597	697
4		Trees	80	20	1144	1244
5		Soil	80	20	1142	1242
6		Water	80	20	225	325
7		Residential	80	20	1168	1268
8		Commercial	80	20	1144	1244
9		Road	80	20	1152	1252
10		Highway	80	20	1127	1227
11		Railway	80	20	1135	1235
12		Parking lot1	80	20	1134	1234
13		Parking lot2	80	20	369	469
14		Tennis court	80	20	328	428
15		Running track	80	20	560	660
Tot.			1200	300	13511	15011

(c)

Fig. 11. University of Houston data set. (a) False-color image. (b) Ground truth. (c) Category and sample settings.

TABLE I

PARAMETER DETAILS OF THE PROPOSED FULLYCONTNET

Part	Layer	Parameter	GN & ReLU
Backbone	Conv1	64, 3, 1, 1, 1	✔
	Average Pool	64, 2, 2, 0, 1	
	Conv2	112, 3, 1, 1, 1	✔
	Conv3	160, 3, 1, 1, 1	✔
	Conv4	208, 3, 1, 1, 1	✔
	Conv5	256, 3, 1, 1, 1	✔
Pyramid-FCM	Conv0	64, 1, 1, 0, 1	
	Conv[1]	16, 1, 1, 0, 1	
	Conv-DR1	128, 1, 1, 0, 1	✔
	Conv-DR2	N[2], 1, 1, 0, 1	
ASP-FCM	Conv0	64, 1, 1, 0, 1	
	Conv1	64, 3, 1, 1, 1	✔
	Dilated Conv1	64, 3, 1, d_1, d_1	✔
	Dilated Conv2	64, 3, 1, d_2, d_2	✔
	Dilated Conv3	64, 3, 1, d_3, d_3	✔
	Conv[1]	16, 1, 1, 0, 1	
	Conv-DR1	128, 1, 1, 0, 1	✔
	Conv-DR2	N[2], 1, 1, 0, 1	

[1] The convolutional layers are conducted to generate keys, queries and values in PAM and the proposed SAM.
[2] Number of categories.

The stochastic gradient descent (SGD) with momentum is used for optimizing, where the momentum is set to 0.9 and the weight decay is configured as 0.0001. All networks adopt the cross-entropy loss. During training, the batch size is always set to 1 because the whole image is directly fed into the network. The training iteration number is set to 1K for all data sets. Following PSPNet [42], the numbers of grid pooling in Pyramid-FCM are separately set as $(1^2, 2^2, 3^2, 6^2)$. The details of the FullyContNet are presented in Table I, and the parameters are depicted as the form of (CN, KS, ST, PS, DR), which separately represents the "channel number," "kernel size," "stride," "padding size," and "dilation rate" of the corresponding layer. In addition, the table also indicates whether the GN and ReLU are used after the layer. There are no data augmentation strategies employing in the proposed methods.

All experiments are repeatedly conducted ten times, where the training samples are randomly selected at the start time and then fixed in each trial, while the remainder makes up the validation set and testing set. In the experiments, all

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TABLE II

OA (%) OF THE VARIOUS SETTING IN SAM AND FCM ON DIFFERENT DATA SETS, WHERE THE FCM USES THE P-C-S SCHEME

Method	PAM & CAM	SAM	Indian Pines	Salinas Valley	Pavia University	University of Houston
Baseline (only FCN)			95.47	98.18	96.13	97.92
Backbone-PP			96.06(+0.59)	99.09(+0.91)	97.65(+1.52)	98.52(+0.60)
FullyContNet-Pyramid	✔		96.88(+1.41)	99.15(+0.97)	98.16(+2.03)	98.67(+0.75)
FullyContNet-Pyramid		✔	96.33(+0.86)	99.16(+0.98)	97.76(+1.63)	98.54(+0.62)
FullyContNet-Pyramid	✔	✔	97.46(+1.99)	99.29(+1.11)	98.71(+2.58)	98.68(+0.76)
Backbone-ASPP			96.07(+0.60)	99.50(+1.32)	98.05(+1.92)	98.00(+0.08)
FullyContNet-ASP	✔		96.69(+1.22)	99.64(+1.46)	98.88(+2.75)	98.27(+0.35)
FullyContNet-ASP		✔	96.27(+0.80)	99.55(+1.37)	98.61(+2.48)	98.12(+0.20)
FullyContNet-ASP	✔	✔	97.09(+1.62)	99.67(+1.49)	98.92(+2.79)	98.31(+0.39)

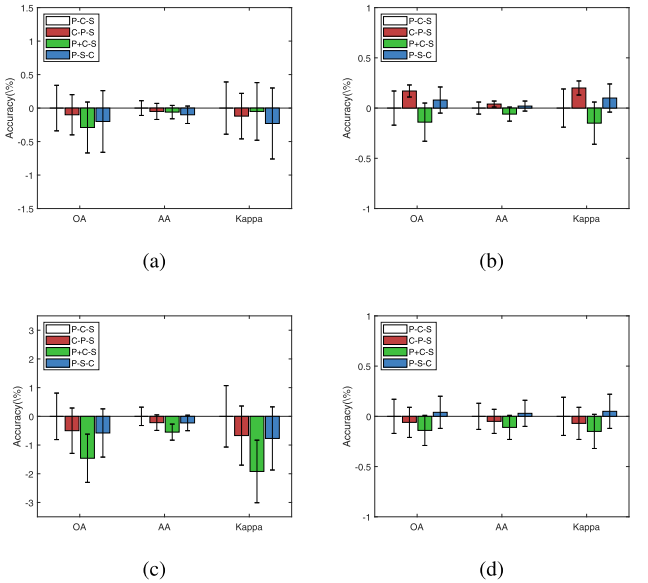

Fig. 12. Comparison of different schemes in Pyramid-FCM on four data sets. (a) Indian Pines. (b) Salinas Valley. (c) Pavia University. (d) University of Houston.

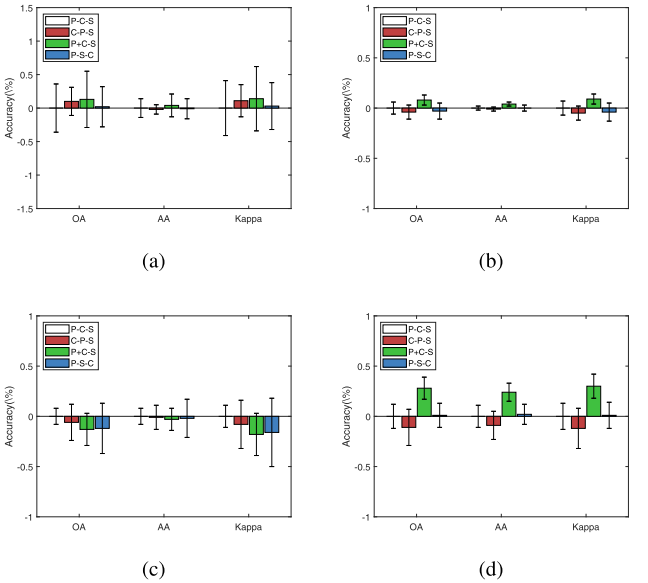

Fig. 13. Comparison of different schemes in ASP-FCM on four data sets. (a) Indian Pines. (b) Salinas Valley. (c) Pavia University. (d) University of Houston.

accuracies are evaluated on the testing set. The validation set is used for best model saving during training. Three evaluation criteria widely using in the hyperspectral remote sensing community are applied in the experiments, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (kappa), which are reported with the mean value and standard deviation. All experiments are implemented on the Intel Xeon Gold 5118 2.30-GHz processor and a single NVIDIA Tesla V100 GPU.

The experiment sequence is arranged as follows. We first conduct the ablation study of SAM and FCM. Then, we compare the four schemes of the proposed Pyramid-FCM and ASP-FCM. The performance comparisons between FullyContNet and other state-of-the-art methods are subsequently presented. In addition, we analyze the stability and complexity of the proposed methods and visualize the learned attention maps for a more comprehensive evaluation.

C. Ablation Study of SAM and FCM

In order to evaluate the effectiveness of the proposed SAM and FCM, a series of ablation experiments are implemented, and the results have been shown in Table II. It can be seen that the multiscale features provided by PP or ASPP modules are beneficial to the classification of the network. In addition, we can find that individually adding PAM is useful than only adopting SAM since the spatial contextual information within features plays a major role in hyperspectral scene parsing. Nevertheless, the accuracies are further improved with the proposed SAM since it can capture the dependencies and build the connections of the multiscale features so as to emphasize the different scale importance of a specific region. When the FCM that simultaneously possesses the PAM and SAM is adopted, the baseline is improved effectively and, finally, achieves the best because each pixel can perceive the contextual information of other pixels regardless that they are in any scales and positions. In addition, the feature representations are also improved using CAM by reweighting the channels. It should note that the effectiveness of the proposed FCM is only demonstrated in the "P-C-S" scheme. The comparison of the four schemes will be presented in later experiments.

D. Comparison of the Schemes in FCM

Figs. 12 and 13 display the comparison of four fully contextual schemes in Pyramid-FCM or ASP-FCM. In order to show the results more clearly, we set the "P-C-S" scheme as the benchmark; thus, the outcomes of other schemes are obtained by subtracting the accuracies of the "P-C-S" scheme.

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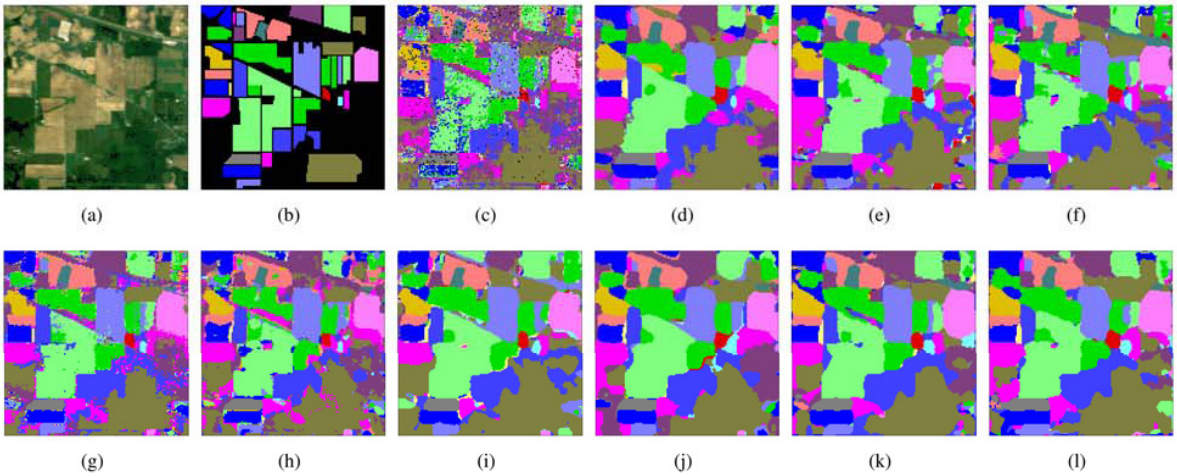

Fig. 14. Parsing result maps on Indian Pines scene. (a) False-color image. (b) Ground truth. (c) SVM. (d) Spectral–spatial unified network (SSUN). (e) Spectral–spatial residual network (SSRN). (f) Pyramid-CNN. (g) Adaptive spectral–spatial multiscale network (ASSMN). (h) SSFCN. (i) ENL-FCN. (j) FreeNet. (k) FullyContNet-Pyramid. (l) FullyContNet-ASP.

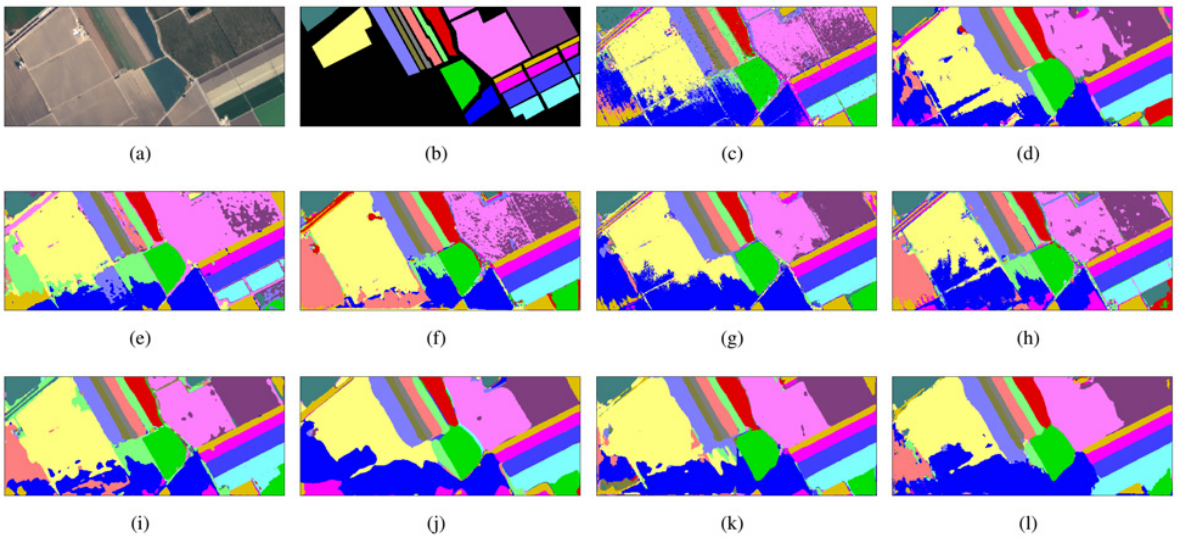

Fig. 15. Parsing result maps on Salinas Valley scene. (a) False-color image. (b) Ground truth. (c) SVM. (d) SSUN. (e) SSRN. (f) Pyramid-CNN. (g) ASSMN. (h) SSFCN. (i) ENL-FCN. (j) FreeNet. (k) FullyContNet-Pyramid. (l) FullyContNet-ASP.

If the results are positive, meaning that the current scheme performs well than the "P-C-S" scheme, the negative values are the opposite. Note that we only minus the average values of ten trials, while the values of standard deviation are not affected, which has been presented on the figures as the error bar.

The OA, AA, and kappa have a similar trend, making it convenient for us to choose the most suitable scheme. In Fig. 12, the "P-C-S" scheme performs the best on two scenes, including Indian Pines and Pavia University, while the "C-P-S" and "P-S-C" schemes separately achieve the highest accuracy on Salina Valley and University of Houston images. However, the accuracy gaps of different schemes on Salina Valley and University of Houston data sets are relatively small compared with other images, such as the Pavia University scene. Thus, we adopt the "P-C-S" scheme in the FullyContNet-Pyramid at the later experiments. According to the same principle, as shown in Fig. 13, the "P + C-S" scheme is selected to implement the FullyContNet-ASP because it takes the lead on all scenes except Pavia University.

TABLE III
COMPARISON OF DIFFERENT METHODS ON INDIAN PINES SCENE (%)

Level	Method	OA	AA	Kappa
Pixel	SVM [16]	77.24 ±0.85	85.22 ±1.37	74.13 ±0.94
Patch	SSUN [30]	95.78 ±0.91	98.04 ±0.38	95.14 ±1.04
	SSRN [27]	95.70 ±1.51	95.30 ±1.90	95.05 ±1.72
	Pyramid-CNN [28]	94.92 ±0.80	97.80 ±0.33	94.15 ±0.91
	ASSMN [31]	91.10 ±3.59	95.15 ±2.50	89.80 ±4.06
Image	SSFCN [34]	86.74 ±0.82	92.87 ±0.94	84.81 ±0.93
	ENL-FCN [38]	95.28 ±1.02	97.75 ±0.41	94.54 ±1.17
	FreeNet [35]	96.76 ±0.28	**98.80** ±**0.10**	96.26 ±0.32
	FullyContNet-Pyramid (Ours)	**97.46** ±**0.34**	98.66 ±0.11	**97.05** ±**0.39**
	FullyContNet-ASP (Ours)	97.31 ±0.42	98.75 ±0.17	96.88 ±0.48

E. Compare With the State of the Art

Tables III–VI quantitatively demonstrate the performance comparison of the proposed methods with other

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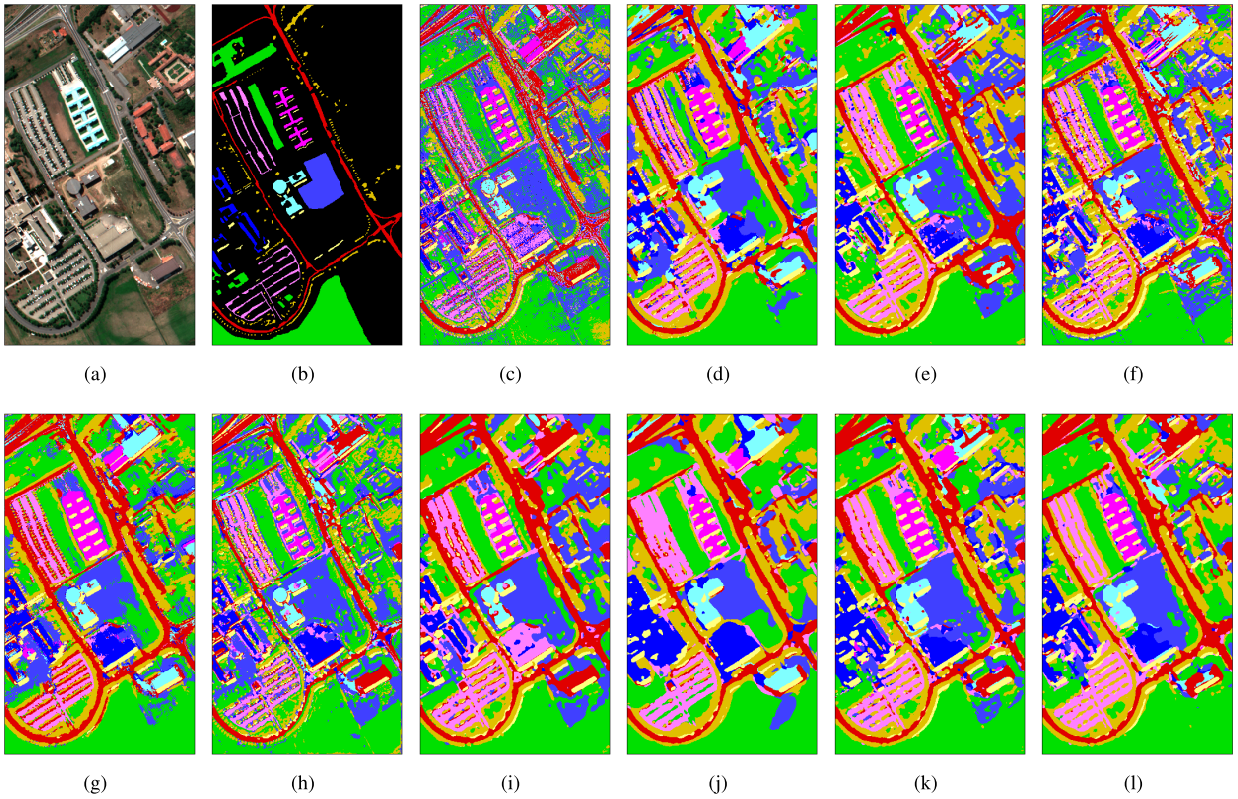

Fig. 16. Parsing result maps on the Pavia University scene. (a) False-color image. (b) Ground truth. (c) SVM. (d) SSUN. (e) SSRN. (f) Pyramid-CNN. (g) ASSMN. (h) SSFCN. (i) ENL-FCN. (j) FreeNet. (k) FullyContNet-Pyramid. (l) FullyContNet-ASP.

TABLE IV
COMPARISON OF DIFFERENT METHODS ON SALINAS VALLEY SCENE (%)

Level	Method	OA	AA	Kappa
Pixel	SVM [16]	90.64 ±0.54	95.61 ±0.22	89.56 ±0.59
Patch	SSUN [30]	97.10 ±0.45	98.83 ±0.15	96.77 ±0.50
	SSRN [27]	94.12 ±1.30	97.42 ±0.46	93.46 ±1.43
	Pyramid-CNN [28]	92.30 ±0.53	97.13 ±0.18	91.44 ±0.59
	ASSMN [31]	96.49 ±1.27	98.62 ±0.50	96.09 ±1.41
Image	SSFCN [34]	95.20 ±0.32	97.70 ±0.19	94.65 ±0.36
	ENL-FCN [38]	97.93 ±0.36	99.08 ±0.15	97.69 ±0.40
	FreeNet [35]	99.64 ±0.16	99.83 ±0.06	99.59 ±0.18
	FullyContNet-Pyramid (Ours)	99.29 ±0.17	99.72 ±0.06	99.20 ±0.19
	FullyContNet-ASP (Ours)	**99.75 ±0.05**	**99.88 ±0.02**	**99.72 ±0.05**

TABLE V
COMPARISON OF DIFFERENT METHODS ON PAVIA UNIVERSITY SCENE (%)

Level	Method	OA	AA	Kappa
Pixel	SVM [16]	86.60 ±0.92	89.61 ±0.38	82.56 ±1.14
Patch	SSUN [30]	93.75 ±1.44	95.73 ±0.72	91.79 ±1.86
	SSRN [27]	98.22 ±0.77	97.84 ±1.04	97.63 ±1.01
	Pyramid-CNN [28]	94.32 ±1.02	95.35 ±1.40	92.49 ±1.32
	ASSMN [31]	94.76 ±1.07	96.76 ±0.47	93.10 ±1.38
Image	SSFCN [34]	89.91 ±0.68	93.60 ±0.52	86.84 ±0.88
	ENL-FCN [38]	97.58 ±1.08	98.10 ±0.83	96.79 ±1.43
	FreeNet [35]	98.39 ±1.13	98.83 ±0.65	97.85 ±1.49
	FullyContNet-Pyramid (Ours)	98.71 ±0.68	**99.19 ±0.34**	98.28 ±0.91
	FullyContNet-ASP (Ours)	**98.79 ±0.16**	98.99 ±0.11	**98.38 ±0.21**

state-of-the-art approaches. The SVM, which directly regards the raw spectral vector as the feature and fed it into the classifier, has been selected as the representation of the pixel-level methods. For the patch-level methods, the SSRN [27] and Pyramid-CNN [28], which exploit the spectral–spatial information using 3-D convolution or the designed skip-connection, and the SSUN [30] and ASSMN [31], which separately capture the channel or position contexts with LSTM or convolutional LSTM, are simultaneously taking part in the performance comparison. Note that the image-level methods have been introduced in Section II-B. In our experiments, the SVM is implemented with the radial basis function kernel. Following [29], two important hyper factors, C and γ, are separately determined through fivefold cross-validation with the grid search in the range of $\{2^{-5}, 2^{-4}, \ldots, 2^{19}\}$ and $\{2^{-15}, 2^{-14}, \ldots, 2^{5}\}$ using the LIBSVM [65], while all patch- and image-level approaches are conducted with the settings in the corresponding original literature. Nevertheless, the SVM gets the worst accuracies since it only adopts the spectral information. The classification is improved by the patch-level methods with the introduction of spatial information. However, the limited size of the input data restricts the improvement of

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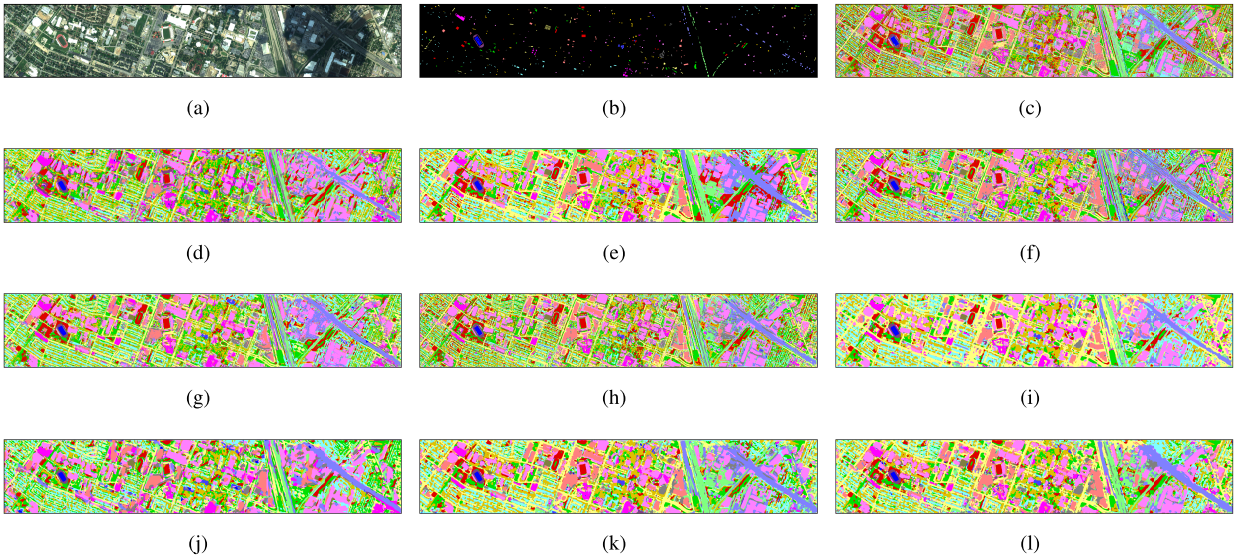

Fig. 17. Parsing result maps on the University of Houston scene. (a) False-color image. (b) Ground truth. (c) SVM. (d) SSUN. (e) SSRN. (f) Pyramid-CNN. (g) ASSMN. (h) SSFCN. (i) ENL-FCN. (j) FreeNet. (k) FullyContNet-Pyramid. (l) FullyContNet-ASP.

TABLE VI
COMPARISON OF DIFFERENT METHODS ON UNIVERSITY OF HOUSTON SCENE (%)

Level	Method	OA	AA	Kappa
Pixel	SVM [16]	92.97 ±0.54	92.86 ±0.60	92.38 ±0.58
Patch	SSUN [30]	94.78 ±0.57	95.64 ±0.48	94.35 ±0.62
	SSRN [27]	94.69 ±3.94	95.63 ±3.00	94.26 ±4.26
	Pyramid-CNN [28]	96.10 ±0.74	96.74 ±0.61	95.78 ±0.80
	ASSMN [31]	96.41 ±2.20	96.64 ±2.21	96.11 ±2.38
Image	SSFCN [34]	95.47 ±0.38	96.08 ±0.36	95.09 ±0.41
	ENL-FCN [38]	98.18 ±0.30	98.40 ±0.22	98.03 ±0.32
	FreeNet [35]	98.53 ±0.26	98.80 ±0.23	98.41 ±0.28
	FullyContNet-Pyramid (Ours)	**98.68 ±0.17**	**98.96 ±0.13**	**98.57 ±0.19**
	FullyContNet-ASP (Ours)	98.59 ±0.11	98.89 ±0.09	98.47 ±0.12

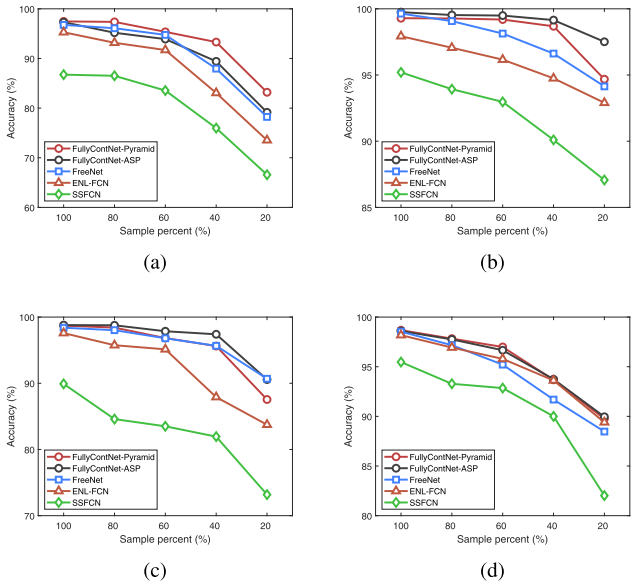

Fig. 18. OA change of the image-level methods when the training samples are continuously reduced on the four scenes. (a) Indian Pines. (b) Salinas Valley. (c) Pavia University. (d) University of Houston.

the classification because the information extraction is still conducted on the patches in a fixed size, meaning that the networks are trapped in a limited receptive field and cannot model the long-range dependence. The image-level methods further improve the accuracies because the networks can process the whole image and possess a much wider vision field by stacking the fully convolutional layers. Specifically, comparing with other approaches, our FullyContNets perform the best since they possess the ability of simultaneously capturing the contextual information lying across spatial positions, scales, and spectral bands with the help of FCM. The contexts obtained with FCM are more generic compared to the simply expanded receptive fields using multiple convolutional layers because they involve the information of other features under the aid of SAM.

Figs. 14–17 present the scene parsing results of the methods, while the corresponding false-color image and ground truth are also given. It is obvious that the pixel-based SVM generates the roughest map due to only considering the spectral information. The object smoothness of patch-level methods locates between the pixel and image levels since the networks consider the surrounding environments of the target position. However, with the limited receptive fields, these methods tend to perform confusion at the complex regions, such as the boundary of objects. In image-level networks, the category of each pixel can be determined using more information because of the larger receptive field. Thus, these methods have less misclassification, producing a high-quality classification map with smooth objects and well-maintained boundaries. In spite of this, there are still a large number of misclassifications in SSFCN and ENL-FCN, such as the middle building in Pavia University scene and the shattered Fallow in Salina Valley

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TABLE VII

OA (%) VERSUS DIFFERENT BACKBONE DEPTHS OF THE PROPOSED METHODS

Depth	5	4	3	2	1
Indian Pines					
FullyContNet-Pyramid	97.46	97.39	96.19	95.89	95.26
FullyContNet-ASP	97.31	96.87	96.66	96.54	96.29
Salinas Valley					
FullyContNet-Pyramid	99.29	99.12	99.10	98.86	98.36
FullyContNet-ASP	99.75	99.57	99.51	99.26	99.17
Pavia University					
FullyContNet-Pyramid	98.71	98.36	97.91	97.66	97.08
FullyContNet-ASP	98.79	98.63	98.40	97.86	97.34
University of Houston					
FullyContNet-Pyramid	98.68	98.46	97.98	97.95	97.80
FullyContNet-ASP	98.59	98.58	98.58	98.27	97.27

scene, while the FreeNet is suffered from the challenge of oversmoothing because it adopts the encoder–decoder architecture, where the features need to upsample to the original size from a relatively low resolution. Compared with the above approaches, the proposed methods take the lead in both surface smoothing and outline preserving. Specifically, benefitting from the only $2\times$ downsampling and the fully contextual capturing, our networks keep the high-resolution parsing and perform well even if at some difficult positions, for example, the discriminative edges and corners of the central purple part in the Indian Pines scene and the well-preserved small trees in Pavia University scene.

As shown earlier, the image-level methods usually perform better than the approaches of pixel and patch levels. To check the stability of these methods, we further evaluate them in the situation of fewer training samples, and the results are shown in Fig. 18, where the "sample percent" is the proportion of training sample number in Figs. 8–11. It can be seen that the accuracies are falling more and more quickly as the labels decreasing. The SSFCN is always on the bottom and far from the remaining methods, while our FullyContNet-Pyramid and FullyContNet-ASP still perform to be competitive and achieve the best regardless of scenes, demonstrating the robustness of the proposed methods.

F. Model Sensitivity and Complexity

To evaluate the sensitivity of the proposed methods, we additionally adopt the backbone network with different depths, and the results are listed in Table VII. It can be obviously seen that accuracies of the proposed FullyContNet-Pyramid and FullyContNet-ASP are all decreasing when using the shorter backbone network as a feature extractor, and it seems that FullyContNet-Pyramid is more severely affected than FullyContNet-ASP. Nevertheless, our methods still can achieve the OA higher than 95% even if the depth is set to 1. The above conclusions demonstrate the importance of high-level features for context capturing and the capability of extracted effective fully contextual information with the proposed FCM.

Table VIII presents the number of parameters (Params) and floating-point operations per second (FLOPs) to assess the complexity of the proposed methods. Compared with the series networks based on the PP module, the models based on the ASPP module have higher complexities since there are four branches in original ASPP are preserved in ASP-FCM, which is directly conducted on the whole feature, while the operations in Pyramid-FCM are separately implemented on the grid regions of the feature. Noted the dimension of input features has been reduced using Conv0 (see Fig. 3 and Table I). In addition, it can be observed that the SAM is lightweight since it only adds extra about 2% parameters and computations on the foundation of PP or ASPP. Compared with the original PP and ASPP modules, the complexities of Pyramid-FCM and ASP-FCM also have slightly increments or even decreased, showing the effectiveness of the proposed methods.

TABLE VIII

MODEL COMPLEXITY OF THE PROPOSED NETWORKS (TAKING THE INDIAN PINES DATA SET AS EXAMPLE)

Method	Params (M)	FLOPs (G)
Baseline (only FCN)	1.13	15.30
+PP	1.26(+0.13, 11.50%)	15.96(+0.66, 4.31%)
+PP+SAM	1.28(+0.15, 13.87%)	16.21(+0.91, 5.95%)
+Pyramid-FCM	1.26(+0.13, 11.50%)	17.30(+2.00, 10.33%)
+ASPP	1.64(+0.51, 45.13%)	20.46(+5.16, 33.73%)
+ASPP+SAM	1.67(+0.54, 47.79%)	20.79(+5.49, 35.88%)
+ASP-FCM	1.37(+0.24, 21.24%)	18.22(+2.92, 19.08%)

TABLE IX

RUNNING TIME OF THE PROPOSED METHODS

Scene	Training time (s)	Testing time (s)
FullyContNet-Pyramid		
Indian Pines	820.06	0.17
Salinas Valley	883.79	0.20
Pavia University	1726.06	0.43
University of Houston	4896.28	1.36
FullyContNet-ASP		
Indian Pines	371.70	0.04
Salinas Valley	487.24	0.08
Pavia University	1386.58	0.35
University of Houston	4505.16	1.20

In order to more comprehensively analyze the proposed methods, we further count the running time on each scene, and the results are shown in Table IX. Compared with the training time that costs hundreds of seconds, the testing time of the networks are much smaller and almost achieve the real-time inferencing, possessing the potentiality of efficient hyperspectral scene parsing when allowing off-line training.

G. Visualization of Attention Map

To better understand the mechanism of FCM, we visualize the learned attention maps of multiscale features $\mathbf{X} = \{X_i \in \mathbb{R}^{C\times H\times W}\}_{i=1}^{N_s}$ in Pyramid-FCM, where N_s is 4. Fig. 19 illustrates the corresponding attention maps of the selected two points P_1 and P_2. A_{ij} is the attention map of the jth feature for the specific points in the ith feature, meaning that the pixel in feature X_i can capture the contextual information in X_j. The depth of color represents the relative importance of the captured contexts. It should emphasize that the values of each summed $A_i = \sum_{j=1}^{N_s} A_{ij}$ equal to 1 since the attentions have been normalized, and we can draw the following conclusions.

1) The first three features mainly capture the contexts of the last feature where the rectangle area is marked with homogenous deep red since the information perception is restricted in the smallest grid, indicating that the pixel preferentially obtains the information of the surrounding environments regardless of the categories.

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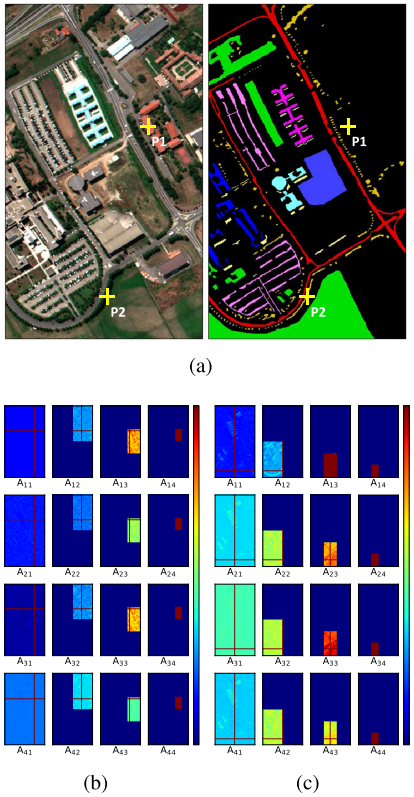

Fig. 19. Learned attention map in Pyramid-FCM on Pavia University scene of two points. (b) House point P_1. (c) Tree point P_2.

2) There are still some detailed information lying in the other features that can be captured by FCM. Concretely, the model will focus on the objects in the same categories as the target pixel when expanding the scope of context capturing. For example, the darker regions in A_{13} and A_{33} of (b) and A_{23}, A_{33}, and A_{43} of (c).
3) With the FCM, the target pixel can also capture the information in a long-distance area to assist classification, such as A_{21}, A_{31}, and A_{41} of the Tree point P_2.

The stronger attentions within feature tend to display in the shape of a criss-cross, where the center is the selected point because we directly employ the RCCA module as the PAM.

V. Conclusion

In this article, we propose the FullyContNet for image-level hyperspectral scene parsing. Different from the traditional pixel- or patch-level approaches that using the local spectral or spatial information, the proposed methods can effectively and adaptively capture the nonlocal context dependencies across spaces, bands, and scales. To this end, we first propose the SAM to obtain the multiscale interfeature dependences. This new paradigm is different from the previous context capturing approaches conducting within features. With the self-attention mechanism, the SAM can adaptively learn the similarity of feature pair and selectively aggregate the multiple features by weighted addition. What is more, when combining the intrafeature information aggregation algorithms, such as PAM and CAM with the SAM on the foundation of PP and ASPP, we can separately acquire the Pyramid-FCM and ASP-FCM to obtain the fully contextual information that simultaneously lies across positions, channels, and features since the SAM can serve as a bridge to associate different features. The parallel paths that contain different grid pooling sizes and dilation rates of atrous convolution in PP and ASPP separately determine the multiple scales of Pyramid-FCM and ASP-FCM. To more effectively obtain the fully contextual information, four schemes are designed and evaluated under the principle that the SAM must conduct after the PAM to achieve better performance as much as possible. The FullyContNet-Pyramid and FullyContNet-ASP are produced by separately placing the Pyramid-FCM and ASP-FCM on the top of the backbone network, which includes five fully convolutional layers, where the only $2\times$ downsampling aims to keep the high-resolution parsing.

A series of qualitative and quantitative comprehensive experiments indicate that the proposed SAM can effectively capture the dependencies and bridge the gaps of multiscale features so as to emphasize the different scale importance of a specific region. The efficiency and capability of FCM have been proved in ablation and complexity analyses. In particular, the nonlocal information capturing patterns of the pixels in Pyramid-FCM are presented with the multiple learned attention maps. The comparisons with the current state-of-the-art approaches on four public scene parsing benchmarks demonstrate that the proposed methods not only achieve the best in the aspect of accuracy but also generate the high-quality classification map that possesses discriminative outlines and continuous surfaces of the objects and can still perform well even with fewer training sample, while the running time evaluation also reveals the real-time inferencing potentiality of the proposed networks.

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